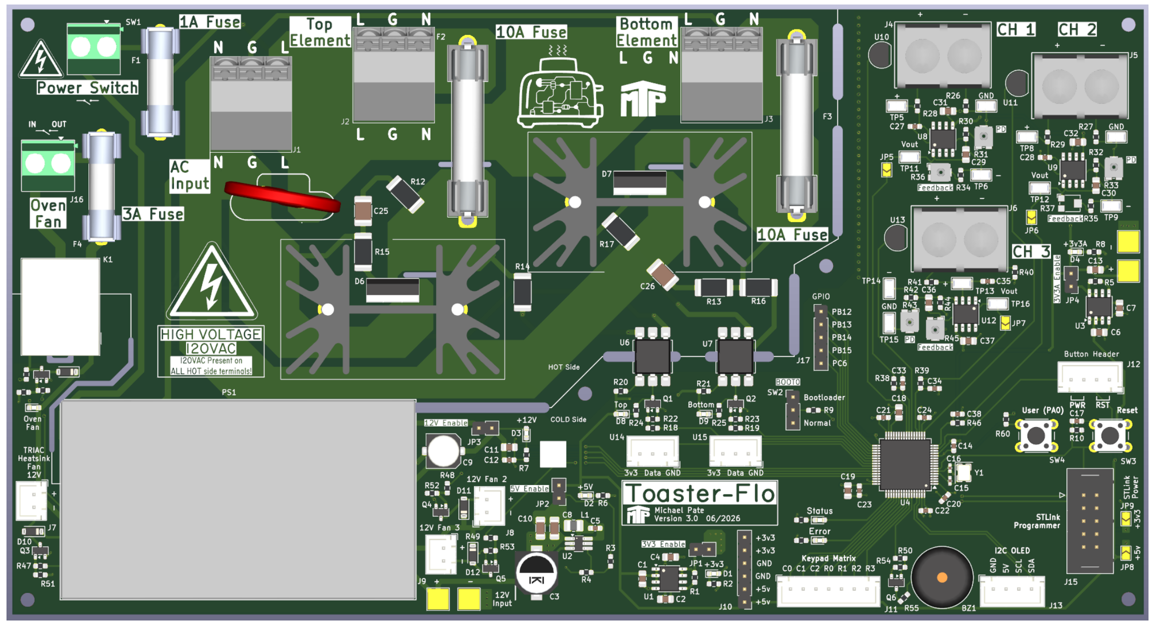
TF Main Board

2026-06-28

Variant:

Rev 2.0

TOP VIEW



DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

NOTES

Add a comment here

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

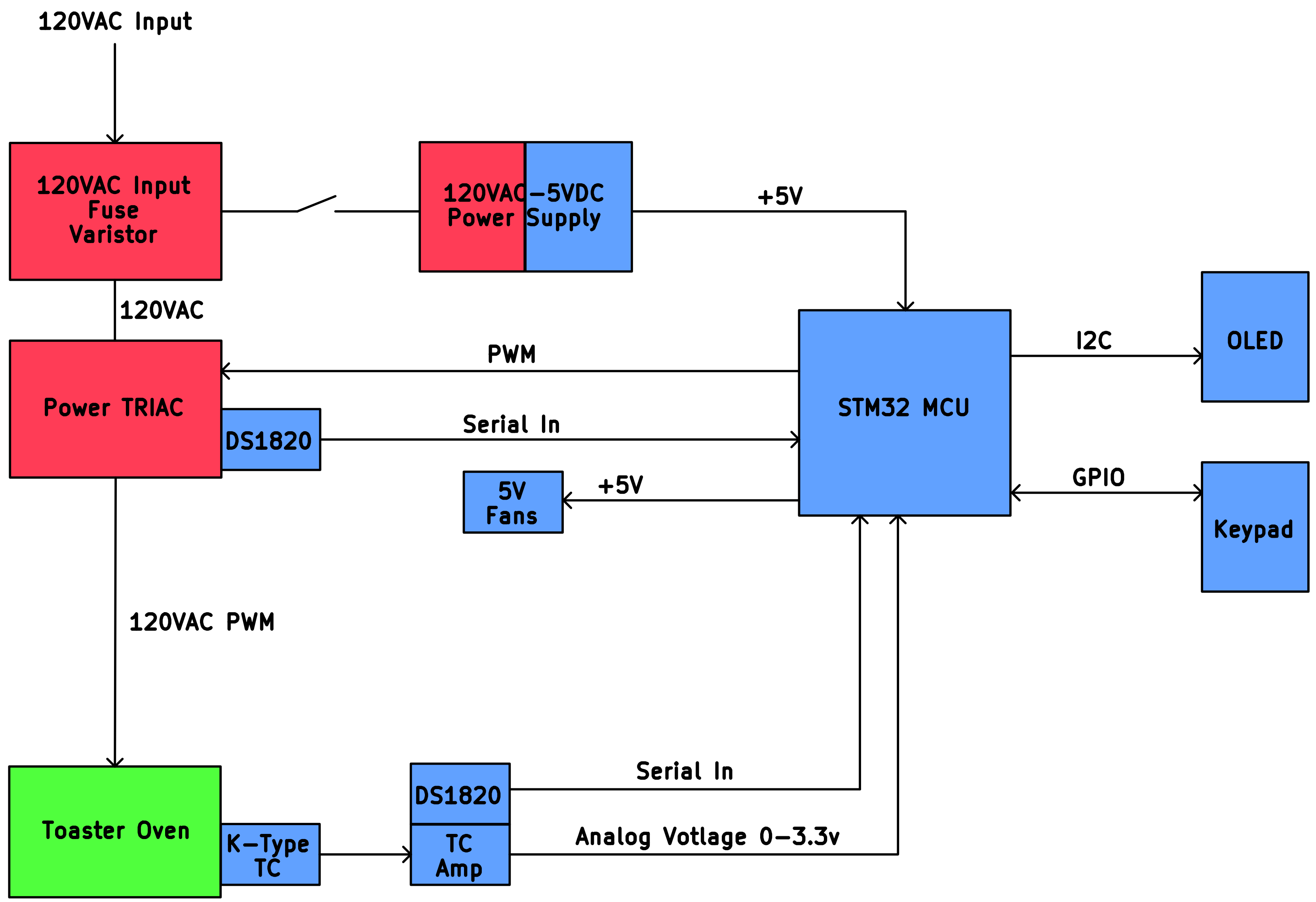
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Release Date

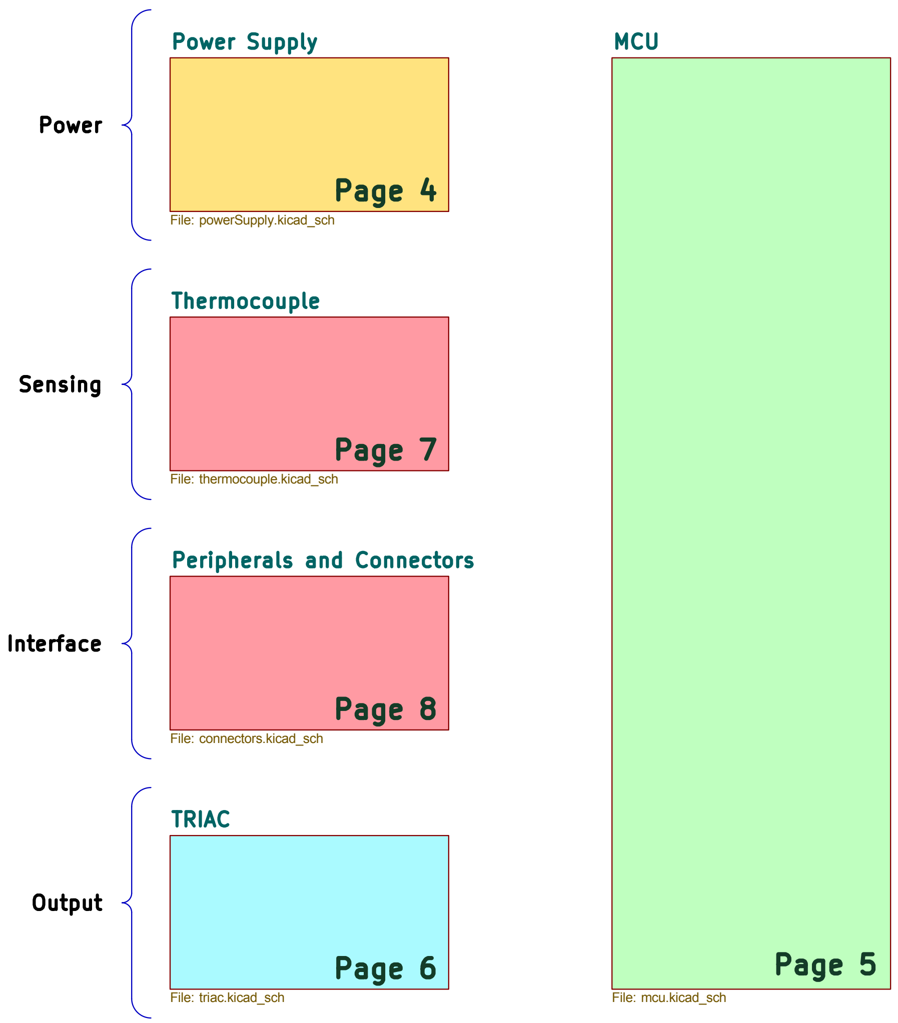
Comments:	Company: MTP Engineering LLC		Variant:
	Board Name: **TF Main Board**		Project Name: **ToasterFlo**
Sheet Title: Cover Page	File Name: mainBoard.kicad_sch	Designer: Michael Pate	Date: Last Modified Date; Revision: 2.0
Sheet Path: /		Reviewer: Michael Pate	Size: A3; Sheet: 1 of 9

[2] Block Diagram



Comments:	Company: MTP Engineering LLC		Variant:	
	Board Name: TF Main Board		Project Name: ToasterFlo	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
Sheet Path: /Block Diagram/		Reviewer: Michael Pate	Size: A3	Sheet: 2 of 9

[3] Project Architecture



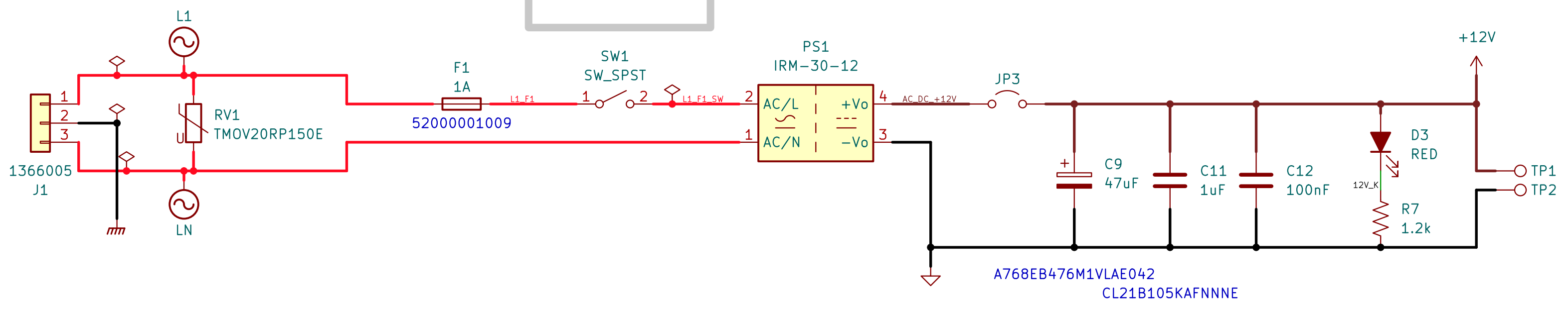
Comments:	Company: MTP Engineering LLC		Variant:	
	Board Name: TF Main Board		Project Name: ToasterFlo	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
Sheet Path: /Project Architecture/		Reviewer: Michael Pate	Size: A3	Sheet: 3 of 9

[4] Power Supply

HIGH VOLTAGE:
Ensure appropriate isolation between AC and DC circuits!

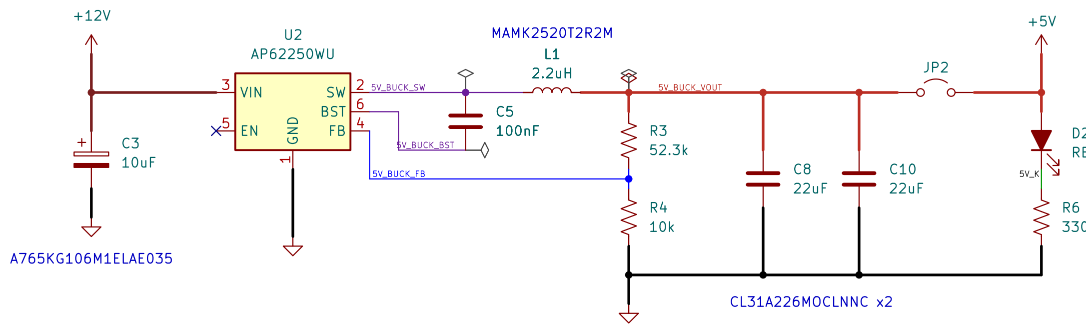
Air isolation gap underneath PS1

SPST Switch
panel-mounted
to enclosure

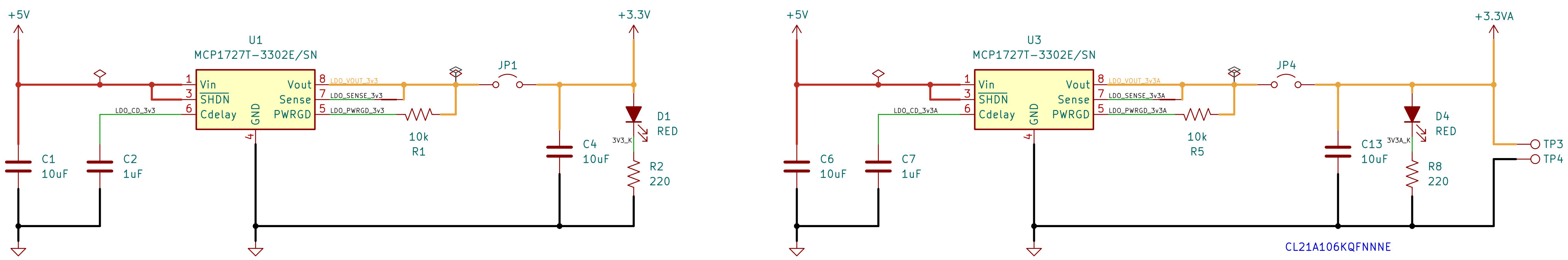


High Frequency! 1.3MHz

Care should be taken to route switching circuit for buck converter



See AP62250 datasheet typical application



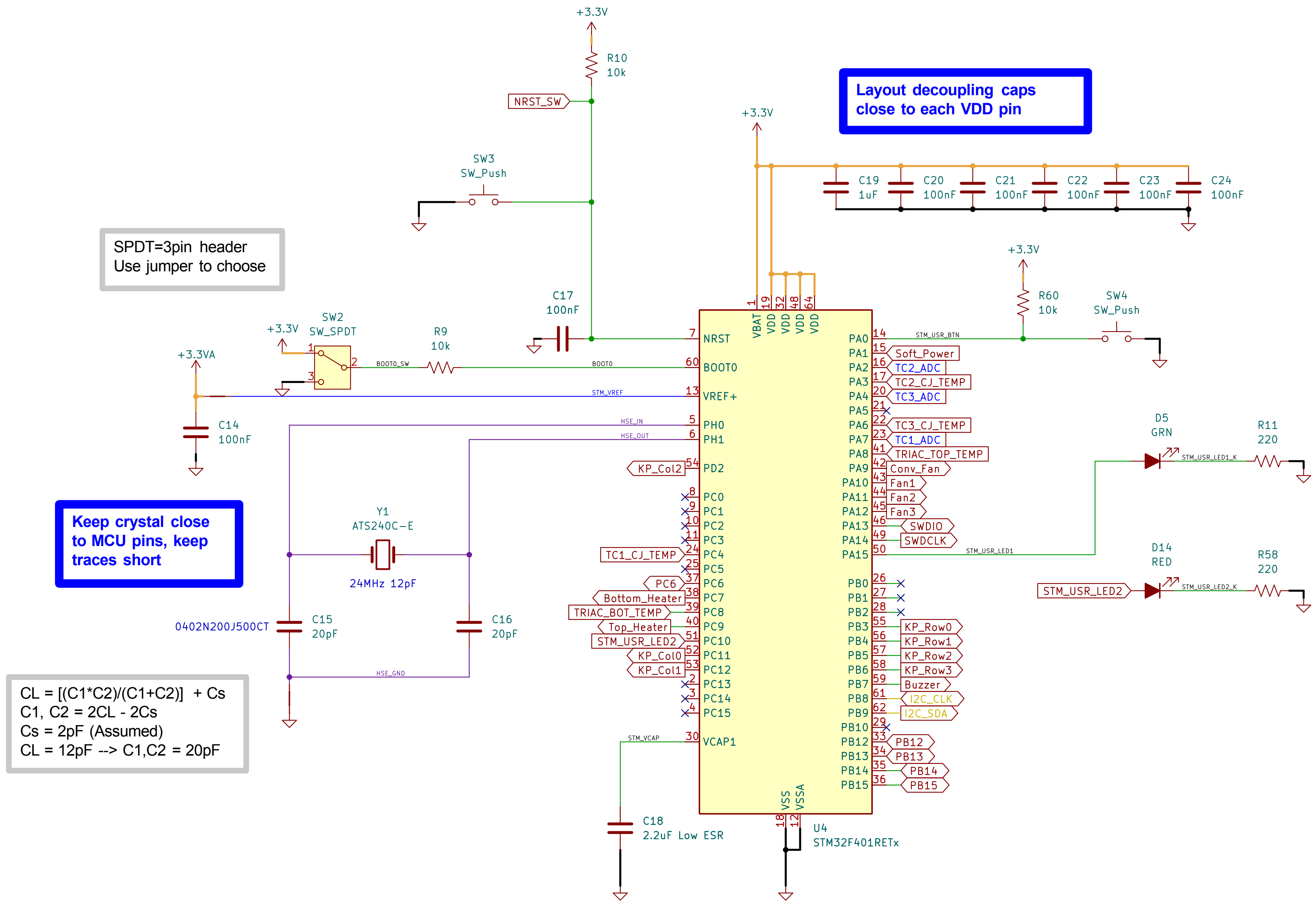
See MCP1727 datasheet typical application

Use M2.5 mounting screws to secure PCB to housing

H1 H2 H3 H4 H5 H6

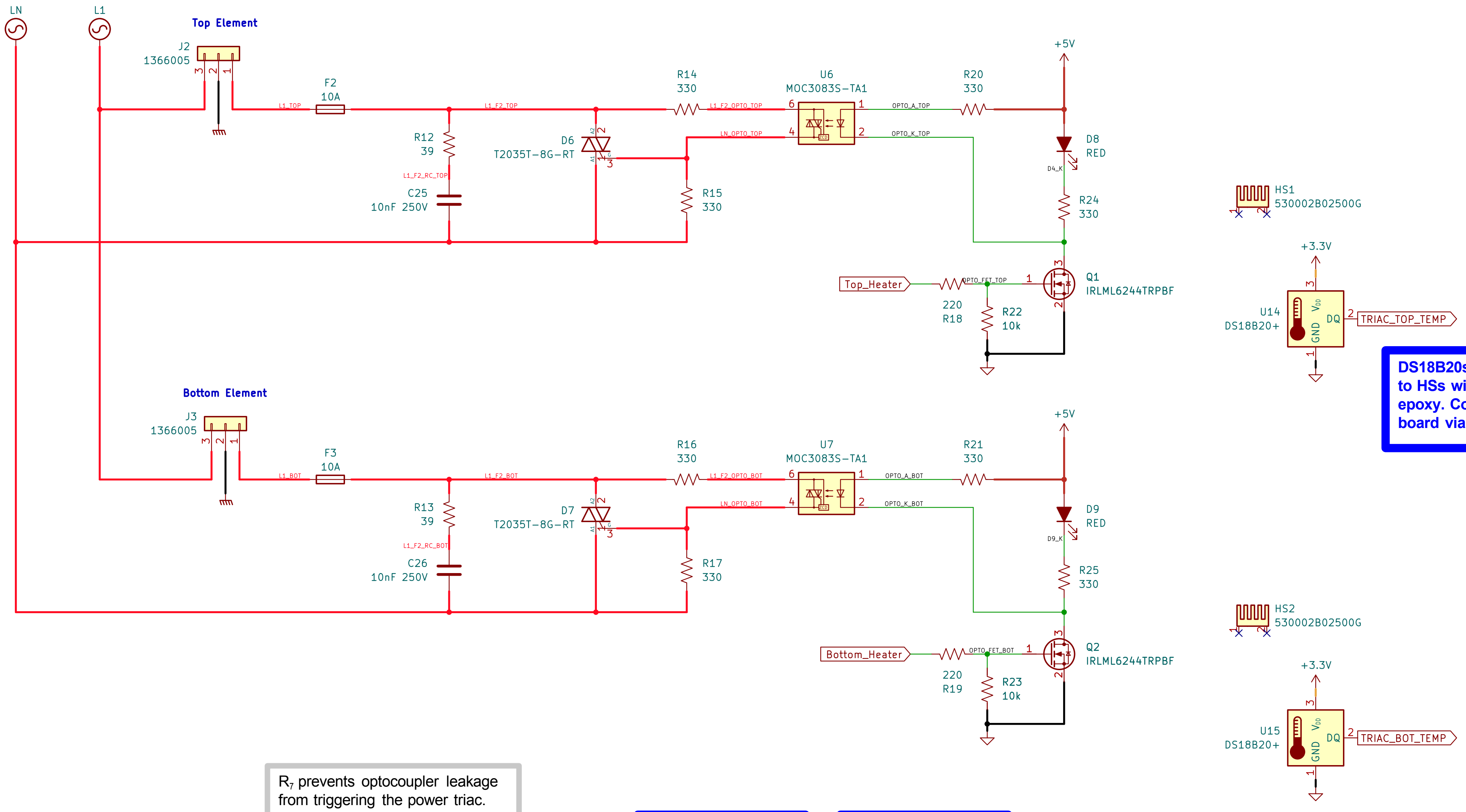
Comments: IRM-10-5 Datasheet; MCP1727T-3302E/SN Datasheet	Company: MTP Engineering LLC			Variant:	
	Board Name: **TF Main Board**			Project Name: **ToasterFlo**	
Sheet Title: Power Supply	File Name: powerSupply.kicad_sch	Designer: Michael Pate		Date: Last Modified Date	Revision: 2.0
Sheet Path: /Project Architecture/Power Supply/		Reviewer: Michael Pate		Size: A3	Sheet: 4 of 9

[5] Microcontroller



Comments:	Company:		Variant:	
IRLML6244TRPBF Datasheet STM32F401RET6 Datasheet	MTP Engineering LLC			
	Board Name: TF Main Board		Project Name: ToasterFlo	
Sheet Title: Microcontroller	File Name: mcu.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
Sheet Path: /Project Architecture/MCU/		Reviewer: Michael Pate	Size: A3	Sheet: 5 of 9

[6] TRIAC Output



DS18B20s mounted to HSs with thermal epoxy. Connected to board via JST XH

Per MOC3085 Datasheet:
$V_{F,LED}$=1.4V @ I_F=20mA
I_{FT}=5mA, $I_{F,MAX}$=50mA

Use R_5=330Ω so I_{LED}=~15mA for 5V IO from MCU

R_7 prevents optocoupler leakage from triggering the power triac.
MOC3083S I_{DRM2}=0.5mA
T2035T I_{GT}=35mA

HIGH VOLTAGE: Ensure appropriate isolation between AC and DC circuits!

HIGH POWER: Ensure appropriate size traces for AC components!

No RC snubber required, see datasheet for T2035T-8G
But for cheap insurance, use a 39Ω/0.01uF RC snubber

At 60Hz, X_{C2}=~265kΩ. At 120V, $I_{C2,RMS}$=~0.45mA

For the MOC3083S, surge current I_{TSM}=1A
$R_{6,MIN}=V_{IN,PK}/1A$
$V_{IN,PK}$=180V for typ. 115VAC supply
$R_{6,MIN}$=180/1=180Ω
Following procedure from AN-3003 pg. 2

StackExchange resources for TRIAC driver design

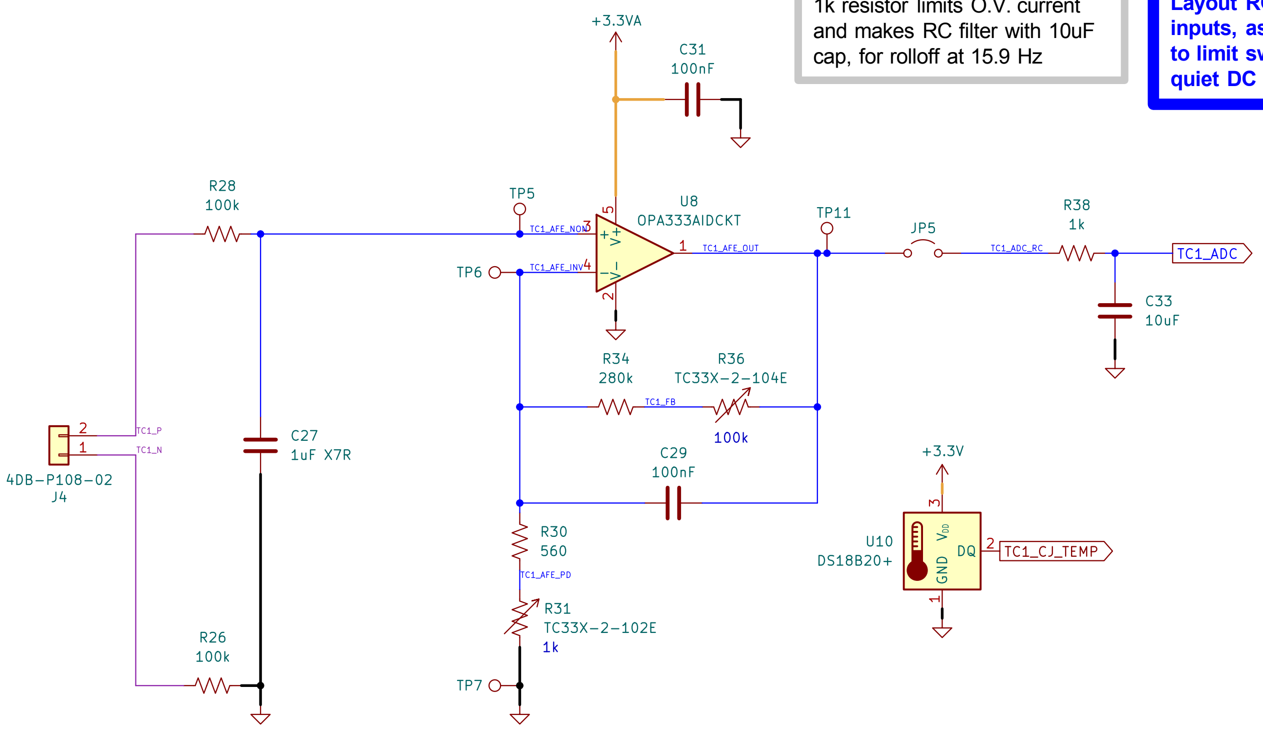
https://electronics.stackexchange.com/questions/381961/triac-optocoupler-circuit
https://electronics.stackexchange.com/questions/248743/how-is-the-gate-trigger-resistor-value-calculated-for-a-triac/248775
https://electronics.stackexchange.com/questions/437809/triac-switching-with-optocoupler
https://electronics.stackexchange.com/questions/53500/optotriactriac-how-do-i-calculate-the-gate-resistor

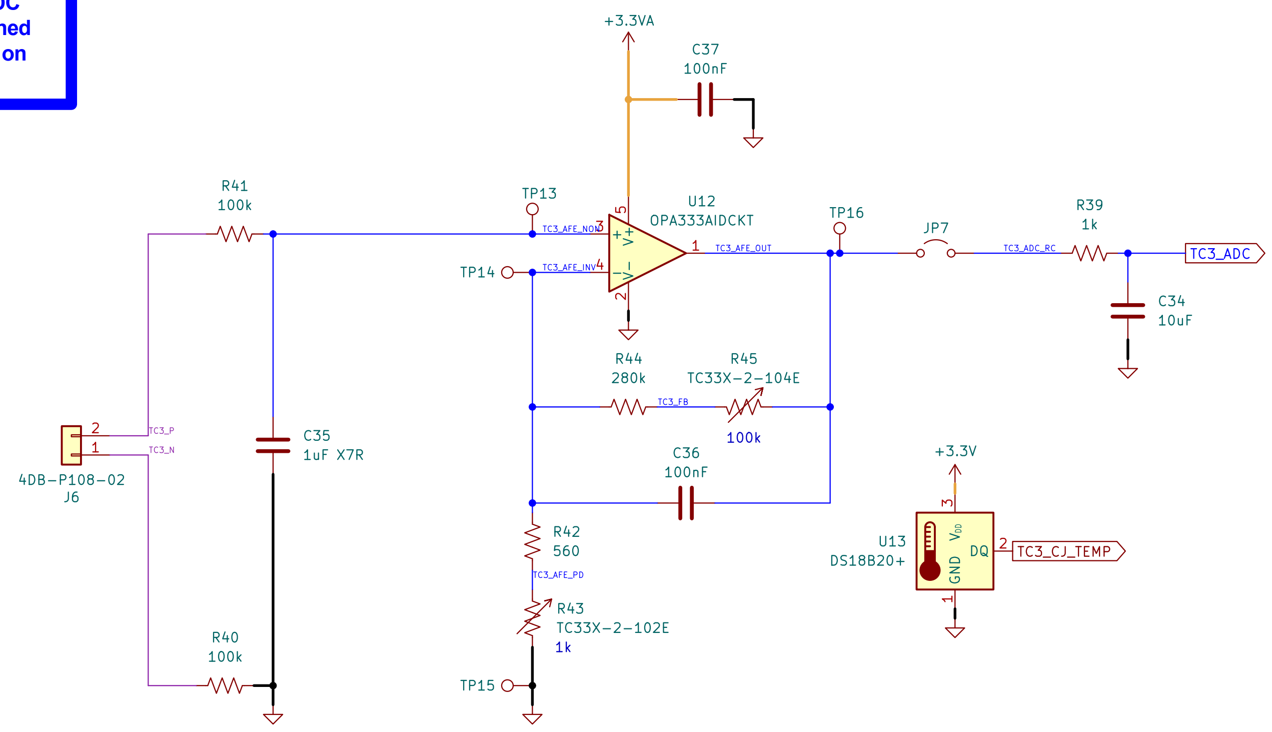
Comments: Link to TRIAC Circuit Example; Link to T2035T Datasheet; Link to MOC3083S Datasheet; Link to Onsemi AN-3003 PDF; CS8674010B0 Heatsink	Company: MTP Engineering LLC		Variant:	
	Board Name: TF Main Board		Project Name: ToasterFlo	
Sheet Title: TRIAC Output	File Name: triac.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
Sheet Path: /Project Architecture/TRIAC/		Reviewer: Michael Pate	Size: A3	Sheet: 6 of 9

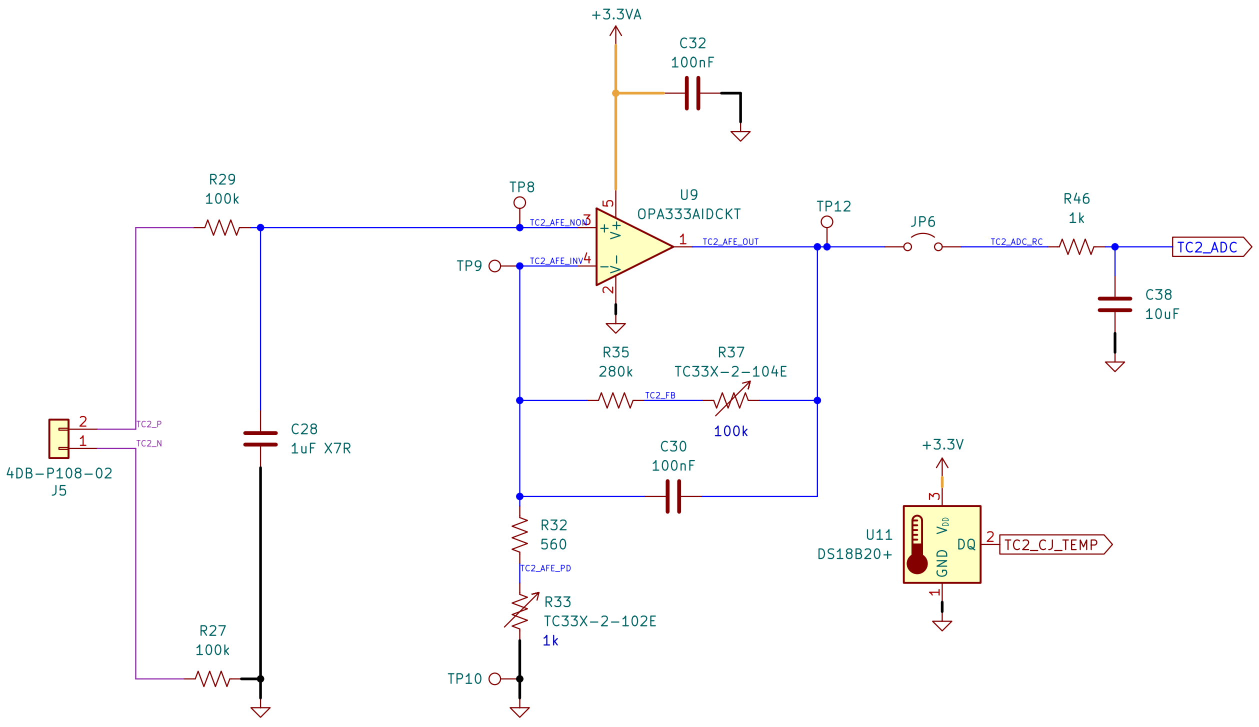
[7] Thermocouple Amplifiers

1k resistor limits O.V. current and makes RC filter with 10uF cap, for rolloff at 15.9 Hz

Layout RC filter near ADC inputs, as filter is designed to limit switching noise on quiet DC signals







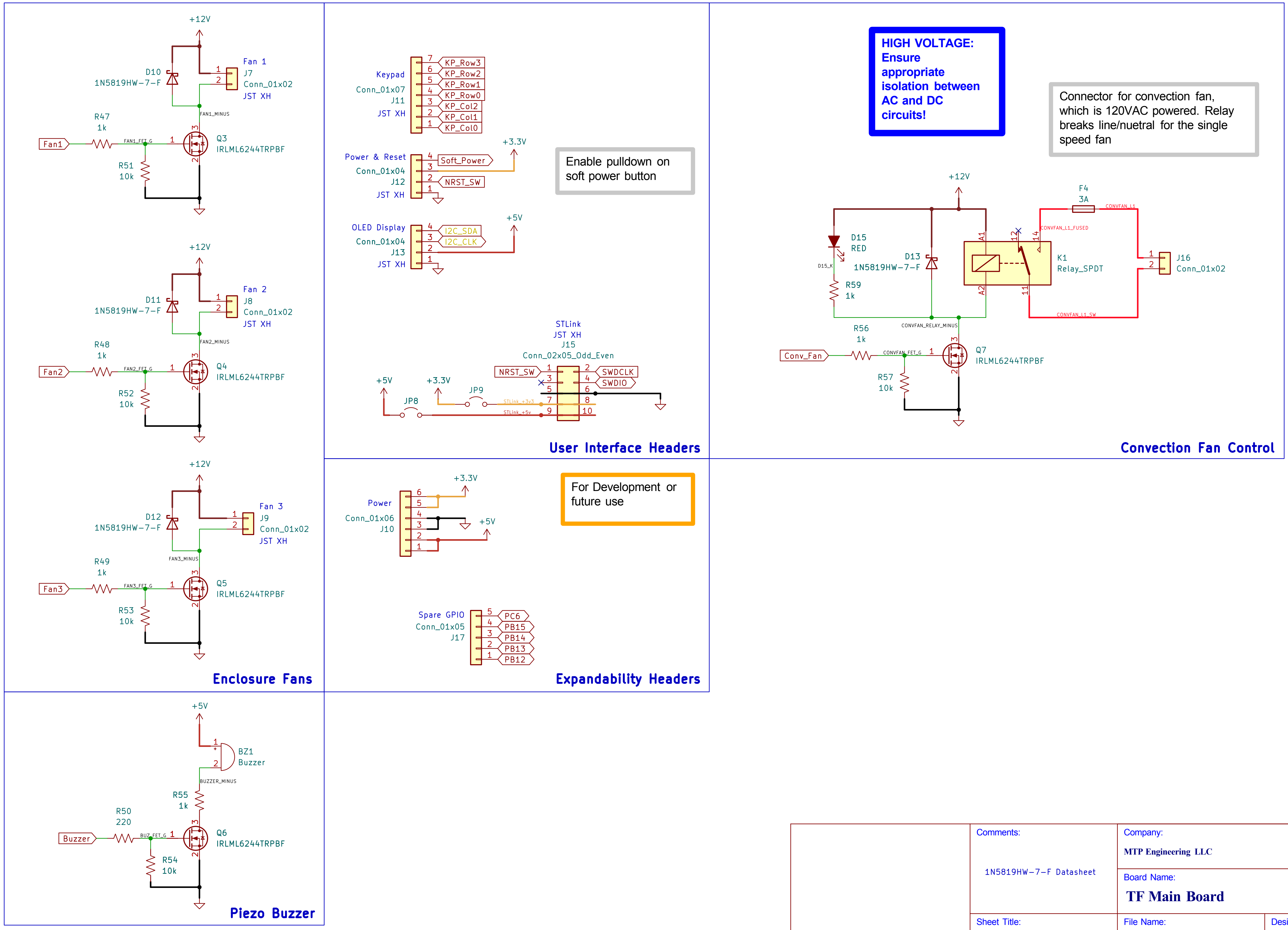
Use RCWCTE testpoint connectors on a Keystone 5019 pad

DS18B20 mounted near TC connector, for cold junction measurement/compensation

RC lowpass filter to remove most AC signals
R = 10kΩ, C = 1 uF
3dB rolloff f = (1 / (2*pi*R*C))
f = (1 / (2*pi*10e3*1e-6)) = 15.92Hz

Comments: K Type TC Voltages; OPA333AIDCKT Datasheet; DS18B20+ Datasheet	Company: MTP Engineering LLC			Variant:	
	Board Name: **TF Main Board**			Project Name: **ToasterFlo**	
Sheet Title: Thermocouple Amplifiers	File Name: thermocouple.kicad_sch	Designer: Michael Pate		Date: Last Modified Date	Revision: 2.0
Sheet Path: /Project Architecture/Thermocouple/		Reviewer: Michael Pate		Size: A3	Sheet: 7 of 9

[8] Interface I/O



Comments: 1N5819HW-7-F Datasheet	Company: MTP Engineering LLC		Variant:	
	Board Name: TF Main Board		Project Name: ToasterFlo	
Sheet Title: Interface I/O	File Name: connectors.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
Sheet Path: /Project Architecture/Peripherals and Connectors/		Reviewer: Michael Pate	Size: A3	Sheet: 8 of 9

[9] Revision History

DD.MM.YYYY - xxx Revision
Variant: xxx

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Comments:	Company: MTP Engineering LLC		Variant:	
	Board Name: TF Main Board		Project Name: ToasterFlo	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
Sheet Path: /Revision History/		Reviewer: Michael Pate	Size: A3	Sheet: 9 of 9